

pod.yml

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PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl apply -f .\pod.yml
pod/frontend-pod created
● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          3s
○ PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> 
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rs.yml

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PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl apply -f .\rs.yml
replicaset.apps/frontend-rs created
● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
frontend-rs   3         3         3       11s
● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          2m52s
frontend-rs-8vmbf  1/1     Running   0          20s
frontend-rs-lcpl8  1/1     Running   0          20s
frontend-rs-ltdpd  1/1     Running   0          20s
○ PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> 
```

deployment.yml

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PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl apply -f .\deployment.yml
deployment.apps/frontend-deployment created
● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deployment  3/3     3            3           21s
● PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
frontend-deployment-9f49ff9c4-5cgg7  1/1     Running   0          33s
frontend-deployment-9f49ff9c4-6qqtq  1/1     Running   0          33s
frontend-deployment-9f49ff9c4-l2g84  1/1     Running   0          33s
frontend-pod  1/1     Running   0          17m
frontend-rs-8vmbf  1/1     Running   0          15m
frontend-rs-lcpl8  1/1     Running   0          15m
frontend-rs-ltdpd  1/1     Running   0          15m
○ PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> 
```

service.yml

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
• PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl apply -f .\service.yml
service/frontend-service created
• PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
calculator-app-service NodePort    10.110.244.73 <none>         80:30000/TCP     14d
demo-app-service    NodePort    10.102.146.151 <none>         80:32000/TCP     7d
frontend-service     NodePort    10.104.205.100 <none>         80:31311/TCP     35s
kubernetes           ClusterIP   10.96.0.1      <none>         443/TCP          21d
○ PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> |
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frontend-service -> v1

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PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
○ PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> minikube service frontend-service

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NAMESPACE	NAME	TARGET PORT	URL
default	frontend-service	80	http://192.168.49.2:31311

```

🔑 Starting tunnel for service frontend-service.

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NAMESPACE	NAME	TARGET PORT	URL
default	frontend-service		http://127.0.0.1:50281

```

🚀 Opening service default/frontend-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
|
```

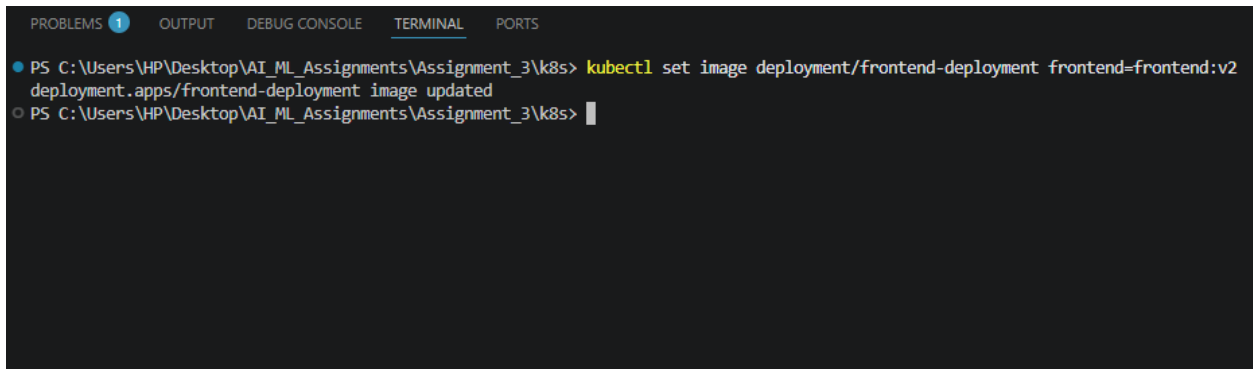


Version 1

Kubernetes Assignment

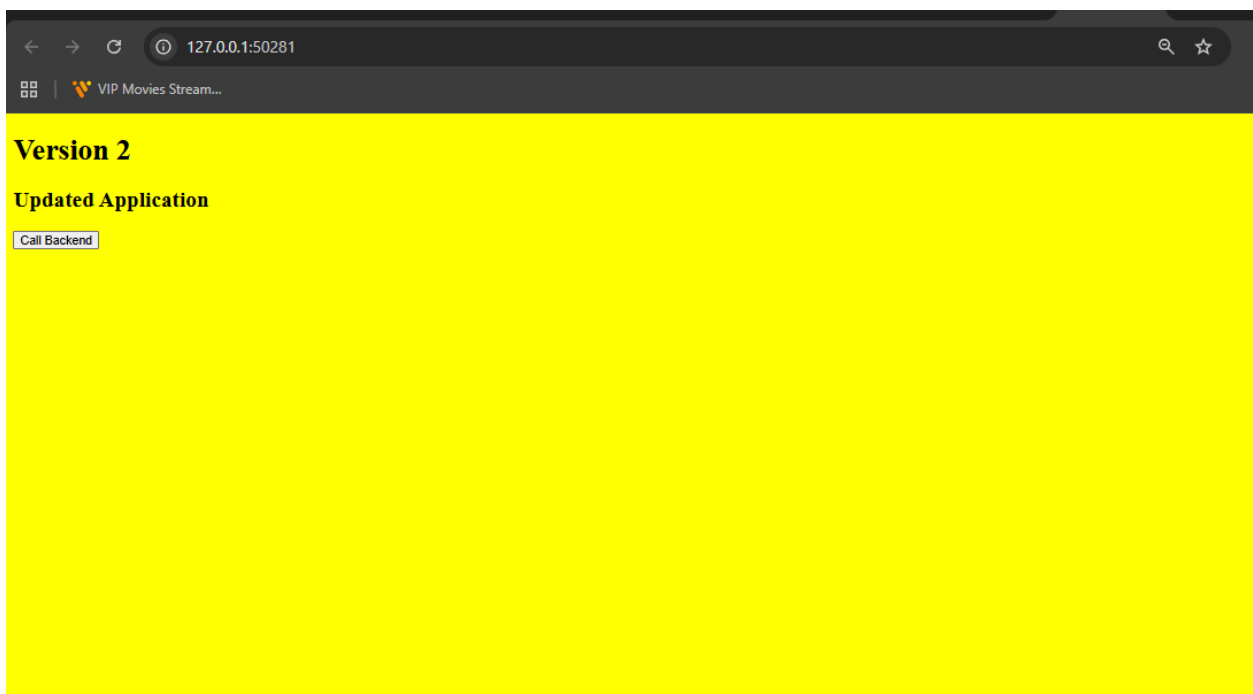
[Call Backend](#)

Rolling Update:



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl set image deployment/frontend-deployment frontend=frontend:v2
deployment.apps/frontend-deployment image updated
PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> |
```

frontend-service ---> updated to v2



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10 |
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\HP\Desktop\AI_ML_Assignments> kubectl get pods -w
frontend-deployment-6764887d77-7xk6j 1/1 Running 0 11s
frontend-deployment-9f49ff9c4-5cgg7 1/1 Terminating 0 20m
frontend-deployment-6764887d77-mgbgc 0/1 Pending 0 0s
frontend-deployment-9f49ff9c4-5cgg7 1/1 Terminating 0 20m
frontend-deployment-6764887d77-mgbgc 0/1 Pending 0 0s
frontend-deployment-6764887d77-mgbgc 0/1 ContainerCreating 0 1s
frontend-deployment-9f49ff9c4-6qqtq 1/1 Terminating 0 20m
frontend-deployment-9f49ff9c4-6qqtq 1/1 Terminating 0 20m
frontend-deployment-9f49ff9c4-5cgg7 0/1 Completed 0 20m
frontend-deployment-9f49ff9c4-6qqtq 0/1 Completed 0 20m
frontend-deployment-6764887d77-mgbgc 0/1 Running 0 2s
frontend-deployment-9f49ff9c4-5cgg7 0/1 Completed 0 20m
frontend-deployment-9f49ff9c4-5cgg7 0/1 Completed 0 20m
frontend-deployment-9f49ff9c4-5cgg7 0/1 Completed 0 20m
frontend-deployment-9f49ff9c4-6qqtq 0/1 Completed 0 20m
frontend-deployment-9f49ff9c4-6qqtq 0/1 Completed 0 20m
frontend-deployment-6764887d77-mgbgc 0/1 Running 0 6s
frontend-deployment-6764887d77-mgbgc 1/1 Running 0 11s
[]
```

Probe verification:

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\HP\Desktop\AI_ML_Assignments\Assignment_3\k8s> kubectl describe pod frontend-deployment-6764887d77-7xk6j
Container ID:   docker://c1a6ae97185200f5e629106b41b4ec9cc979ee2c02858701270a57592aa14799
Image:          frontend:v2
Image ID:       docker://sha256:668647d9782d9c3cd0c7015e17fcfc454d34577cb82e3e91bea0e390ec6a4d67
Port:           80/TCP
Host Port:      0/TCP
State:          Running
  Started:      Sun, 03 May 2026 12:12:57 +0530
Ready:          True
Restart Count:  0
Liveness:       http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
Readiness:      http-get http://:80/ delay=5s timeout=1s period=5s #success=1 #failure=3
Startup:        http-get http://:80/ delay=0s timeout=1s period=5s #success=1 #failure=30
Environment:    <none>
Mounts:
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-hlw1x (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              True
  ContainersReady                    True
  PodScheduled                       True
Volumes:
  kube-api-access-hlw1x:
    Type:  Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    ConfigMapOptional: <nil>
    DownwardAPI: true
QoS Class:  BestEffort
```